



ရန်ကုန်အရှေ့ပိုင်းတက္ကသိုလ်၊ ၂၀၂၆ ပညာသင်နှစ်၊ စာစဉ်အခြေလွှာစာအုပ်

အမည်	- မောင်အောင်ဖြူ
ရံအမှတ်	- ၃-၇၁-၁၆ (၁၂၆)
အထူးပြု	- ဂုဏ်ဗေဒ
Module No./Module Name	- Zool. 2001 / (Invertebrate Zoology)
Assignment I / II	- I

1. According to the site of enzymatic action, chemical digestion is of two types.

(a) Intracellular Digestion

When digestion takes place within the cell, it is called intracellular digestion. The protoplasm of the unicellular animal captures the food in a food vacuole, digests it, discharges the wastes, and incorporates the digested simple organic components into the protoplasm.

(b) Extracellular Digestion

When digestion takes place outside the cell and generally within the digestive tract of the higher animals, it is called extracellular digestion. Digestion is wholly ~~interad~~ intracellular in Protozoa and Porifera. In other phyla, extracellular digestion either supplements the intracellular mechanisms or completely replaces it. In the following discussion, describe the process of feeding and digestion in some invertebrates.

- 2.
- (a) Coxal glands
 - (b) Malpighian tubules
 - (c) Malpighian Tubules

Other arthropods, such as insects, myriapods, and arachnids, have Malpighian tubules, the outgrowths of the alimentary canal, as excretory organs. Malpighian tubules are totally new structures, having no resemblance either to nephridia or coelomoducts. Generally, the Malpighian tubules are composed of single-layered epithelium and are bathed in the blood of the haemocoel. By the process of active secretion, water passes into the lumen

of the Malpighian tubule together with nitrogenous wastes and dissolved salts. The secretion essentially occurs in the distal parts of the tubules. The proximal parts of the tubule as well as the rectum are the sites of absorption. Most of the terrestrial arthropods excrete uric acid.

3. The rotifers are ecologically divided into three groups:
- (a) Alkaline fauna - Found in hard water. Eggs of amictic females can tolerate alkaline conditions.
 - (b) Acid fauna - Found in soft water. Eggs produced by mitotic females can withstand acidic conditions.
 - (c) Transcursion fauna - Some forms produce dormant eggs that can tolerate both alkaline and acidic conditions.



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အမည်

ရံအမှတ်

အထူးပြု

Module No./Module Name

Assignment I / II

- မောင်အောင်ဖြိုး

- ၃-ရခ-၁၆ (၀၂၆)

- ရက် ၆၀၃

- Zool. 2001 / (Invertebrate Zoology)

- II

1. In insects, respiration occurs mainly through an elaborate system of branching and anastomosing tubes which opens to the exterior through pairs of pores, spiracles (usually one pair in each segment) on the lateral sides of the body. These tubes are referred to as tracheae (singular: trachea) and together constitute the tracheal system that delivers oxygen to and removes carbon dioxide from the tissues.

The general plan of the tracheal system of insects is similar to the circulatory system of vertebrates. Tracheal tubes are distributed throughout the body of the insect so that air is brought in direct contact with the tissues and cells of the body.

Tracheal system of insects is classified into the following types based on the number and distribution of functional spiracles:

- (i) Holopneustic - when all the spiracles remain open, e.g., most adults and many larval insects
- (ii) Apneustic - when the spiracles are altogether lacking, e.g., endoparasitic and aquatic larval insects.
- (iii) Hemipneustic - when one or a pair of spiracles remains closed, e.g., some flies, beetles, butterflies, etc.